

Graphing Sine, Cosine, Secant, and Cosecant Functions

Trigonometry Worksheet · Grade 10–12

Name: _____

Date: _____

Learning Objectives

- Identify the amplitude and period of sine and cosine functions
- Graph sine and cosine functions by plotting key points over one period
- Extend understanding to secant and cosecant graphs using their reciprocal relationships

Problems

1. Find the amplitude of the function below:

$$y = 5\sin(\theta)$$

2. Find the period of the function below. Express your answer in degrees.

$$y = \sin(3\theta)$$

3. Find both the amplitude and the period of the function below. Express the period in degrees.

$$y = 4\cos(3\theta)$$

4. Find the period of the function below. Express your answer in degrees.

$$y = 2\sin\left(\frac{\theta}{3}\right)$$

5. List the five key x-values used to partition one period when graphing the function below. Use degrees.

$$y = \sin(3\theta)$$



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6. Describe the starting behavior of the cosine function versus the sine function when graphing one period. Then identify the starting y-value for the function below at theta equals 0.

$$y = 4\cos(3\theta)$$

7. Find the amplitude and period of the function below, then list the five key points (theta, y) for one complete period. Use degrees.

$$y = 2\sin\left(\frac{\theta}{3}\right)$$

8. The function below is a cosecant function. Using the fact that cosecant is the reciprocal of sine, identify where the vertical asymptotes occur within one period. Use degrees.

$$y = \csc(3\theta)$$

9. Find the amplitude and period of the function below. Then describe the full sequence of y-values at the five key points for one period. Use degrees.

$$y = -3\cos(2\theta)$$

10. For the secant function below, find the period, identify the asymptotes, and describe how you would sketch the graph over one period using the corresponding cosine graph as a guide. Use degrees.

$$y = \sec(3\theta)$$



Graphing Sine, Cosine, Secant, and Cosecant Functions — Answer Key

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Answer Key

1. Answer: 5

- The general form is $y = a \sin(b \theta)$
- The amplitude is the absolute value of a
- $|5| = 5$

2. Answer: 120 degrees

- The period formula is $2\pi / b$
- Here $b = 3$, so the period = $2\pi / 3$
- Convert: $2 * 180 / 3 = 360 / 3 = 120$ degrees

3. Answer: Amplitude = 4, Period = 120 degrees

- Amplitude = $|a| = |4| = 4$
- Period = $2\pi / b = 2\pi / 3$
- Convert to degrees: $2 * 180 / 3 = 120$ degrees

4. Answer: 1080 degrees

- $b = 1/3$, so the period = $2\pi / (1/3)$
- Simplify: $2\pi * 3 = 6\pi$
- Convert 6π to degrees: $6 * 180 = 1080$ degrees

5. Answer: 0, 30, 60, 90, 120 degrees

- The period is 120 degrees
- Divide into 4 equal parts: $120 / 4 = 30$ degrees each
- Key points: 0, 30, 60, 90, 120 degrees

6. Answer: Cosine starts at the maximum ($y = 4$); sine starts at the center ($y = 0$)

- Sine starts at the center ($y = 0$) and goes up first
- Cosine starts at the maximum amplitude
- For $y = 4\cos(3\theta)$, at $\theta = 0$: $y = 4 * \cos(0) = 4 * 1 = 4$

7. Answer: Amplitude = 2, Period = 1080 deg; Key points: (0,0), (270,2), (540,0), (810,-2), (1080,0)

- Amplitude = $|2| = 2$
- Period = $2\pi / (1/3) = 6\pi = 1080$ degrees
- Divide 1080 by 4 = 270 degrees per partition
- Key points: (0,0), (270,2), (540,0), (810,-2), (1080,0)

8. Answer: Vertical asymptotes at $\theta = 0, 60$, and 120 degrees



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- $\csc(\theta) = 1/\sin(\theta)$, so asymptotes occur where $\sin(3\theta) = 0$
- $\sin(3\theta) = 0$ when $3\theta = 0, 180, 360$ degrees
- Divide by 3: $\theta = 0, 60, 120$ degrees within one period

9. Answer: Amplitude = 3, Period = 180 deg; Key y-values: -3, 0, 3, 0, -3

- Amplitude = $|-3| = 3$
- Period = $2\pi / 2 = \pi = 180$ degrees
- Cosine normally starts at max, but the negative sign reflects it
- At $\theta = 0$: $y = -3 \cdot \cos(0) = -3$ (starts at minimum)
- Key y-values at $0, 45, 90, 135, 180$ degrees: $-3, 0, 3, 0, -3$

10. Answer: Period = 120 deg; asymptotes at $\theta = 30$ and 90 degrees; secant forms U-shaped curves where cosine reaches its maximum and minimum

- $\sec(\theta) = 1/\cos(\theta)$, so first graph $y = \cos(3\theta)$
 - Period of $\cos(3\theta) = 2\pi/3 = 120$ degrees
 - Asymptotes occur where $\cos(3\theta) = 0$: at $3\theta = 90$ and 270 , so $\theta = 30$ and 90 degrees
 - Where cosine reaches its max (+1), secant opens upward (minimum of 1)
 - Where cosine reaches its min (-1), secant opens downward (maximum of -1)
 - Sketch U-shaped and upside-down U-shaped curves between the asymptotes
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